



```
In [1]: pip install --upgrade seaborn
```

```
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.0.2)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/python3.12/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2026.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
```

```
In [2]: import warnings
warnings.filterwarnings("ignore",category=FutureWarning)
```

```
In [3]: import seaborn as sns
sns.get_dataset_names()
```

```
Out[3]: ['anagrams',  
        'anscombe',  
        'attention',  
        'brain_networks',  
        'car_crashes',  
        'diamonds',  
        'dots',  
        'dowjones',  
        'exercise',  
        'flights',  
        'fmri',  
        'geyser',  
        'glue',  
        'healthexp',  
        'iris',  
        'mpg',  
        'penguins',  
        'planets',  
        'seaice',  
        'taxi',  
        'tips',  
        'titanic']
```

```
In [4]: import matplotlib.pyplot as plt  
        # Load sample dataset  
        tips = sns.load_dataset("tips")  
        # Set visual theme  
        sns.set_theme(style="darkgrid")
```

```
In [5]: tips
```

```
Out[5]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

```
In [6]: # Save to csv
tips.to_csv("tips_dataset.csv", index=False)
```

1. Scatter Plot

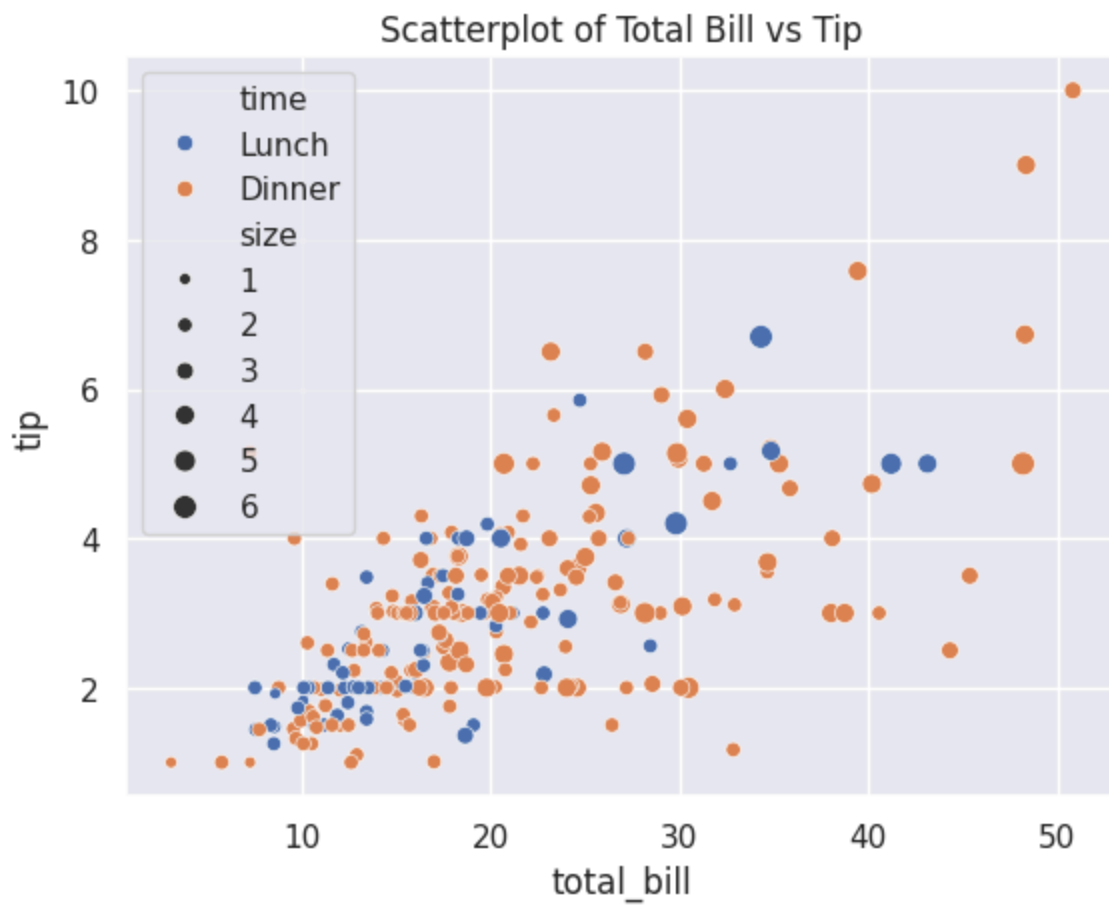
```
In [7]: import os
os.getcwd()
plt.figure(figsize=(8, 6))
```

```
Out[7]: <Figure size 800x600 with 0 Axes>
<Figure size 800x600 with 0 Axes>
```

```
In [8]: sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", size="size",
palette="deep")

plt.title("Scatterplot of Total Bill vs Tip")
```

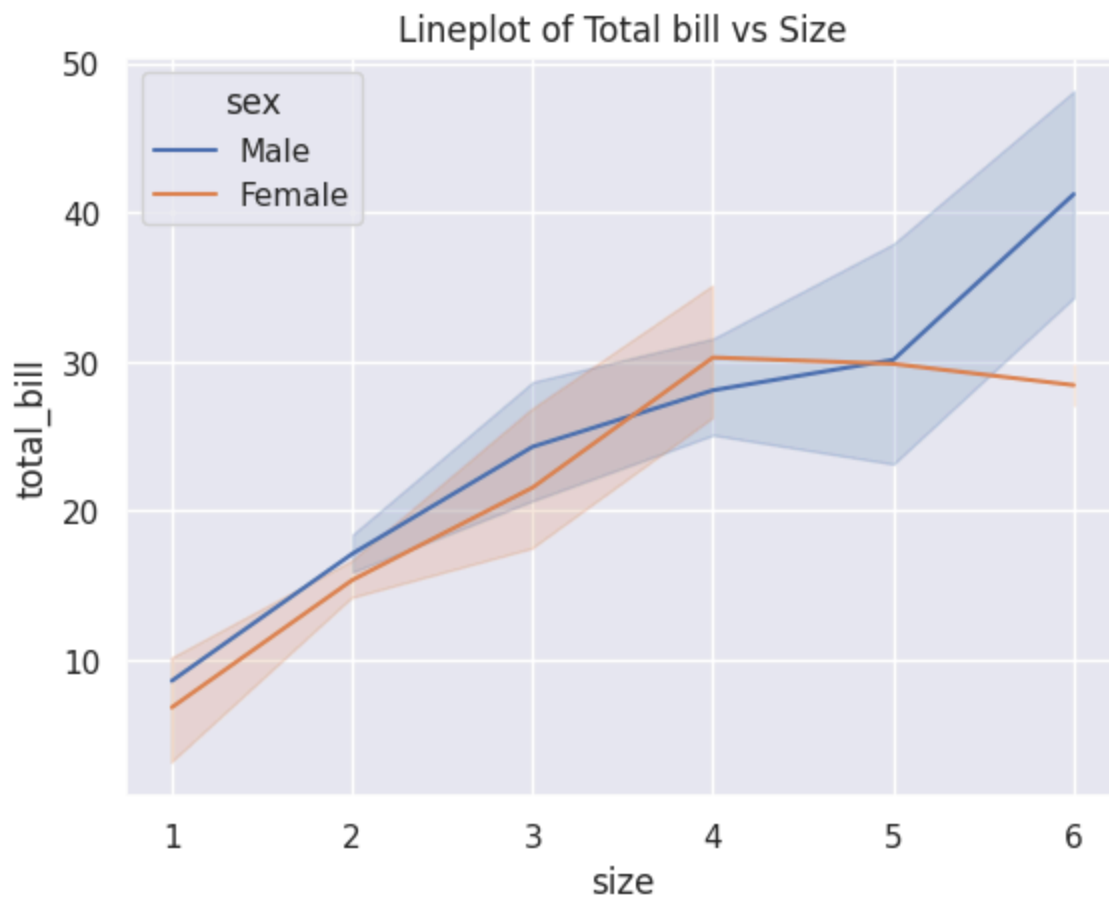
```
Out[8]: Text(0.5, 1.0, 'Scatterplot of Total Bill vs Tip')
```



2. Line Plot

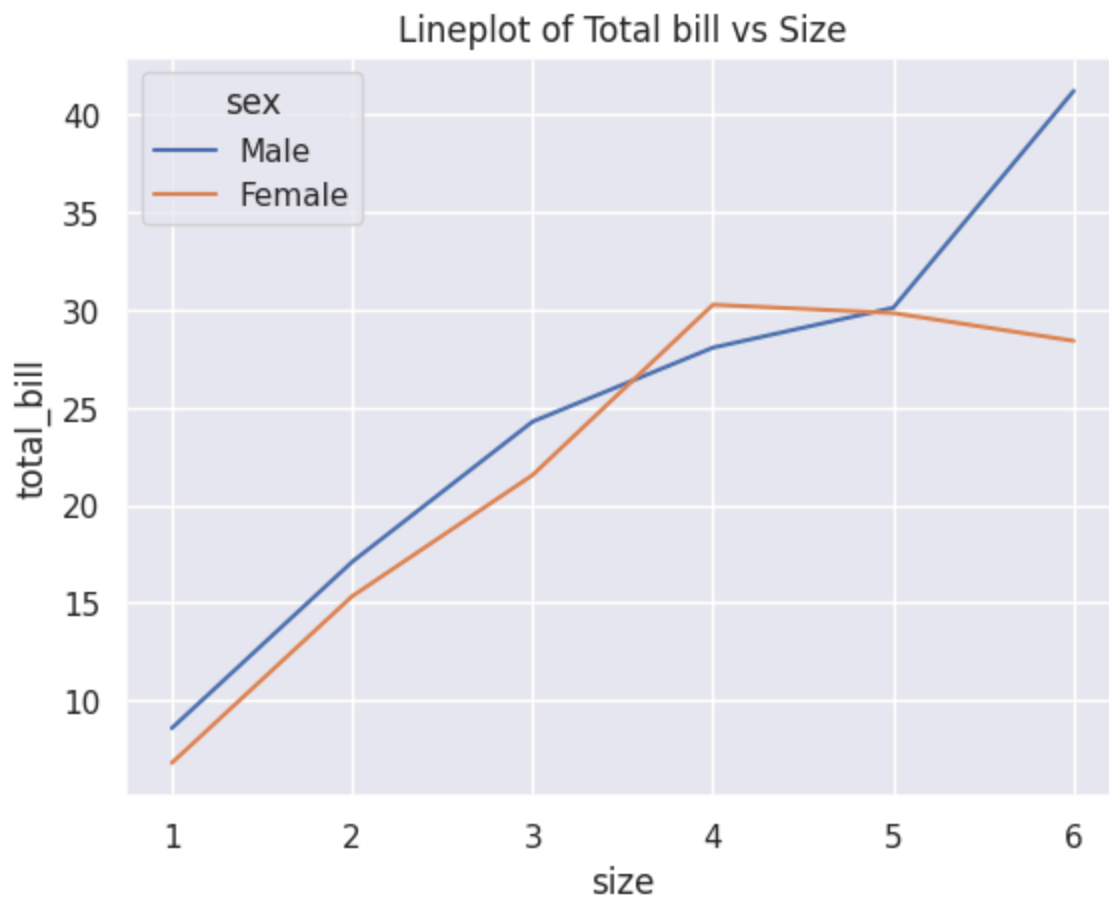
```
In [9]: sns.lineplot(data=tips,x='size',y='total_bill',hue='sex',markers='o')  
plt.title("Lineplot of Total bill vs Size")
```

```
Out[9]: Text(0.5, 1.0, 'Lineplot of Total bill vs Size')
```



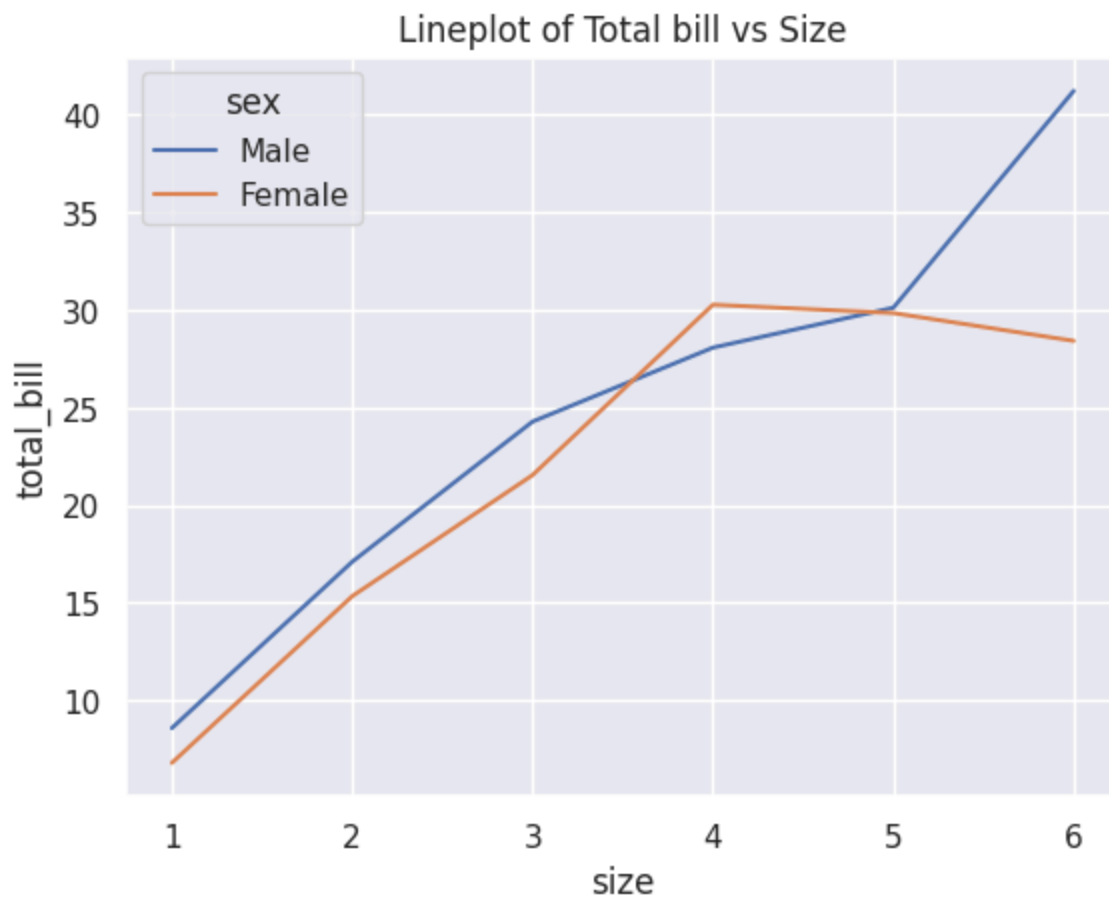
```
In [13]: sns.lineplot(data=tips,x='size',y='total_bill',hue='sex',ci=None,markers='o')  
plt.title("Lineplot of Total bill vs Size")
```

```
Out[13]: Text(0.5, 1.0, 'Lineplot of Total bill vs Size')
```



```
In [10]: sns.lineplot(data=tips,x='size',y='total_bill',hue='sex',ci=None,markers='o')  
plt.title("Lineplot of Total bill vs Size")
```

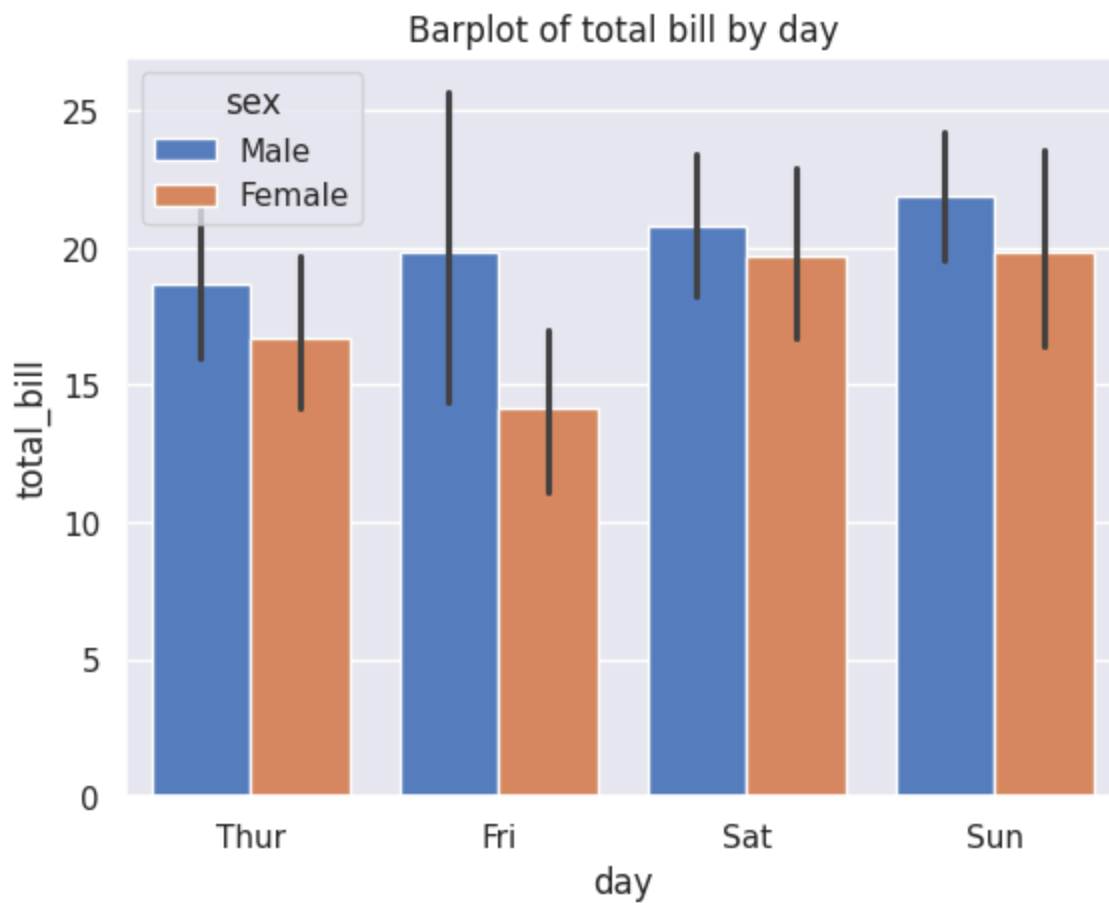
```
Out[10]: Text(0.5, 1.0, 'Lineplot of Total bill vs Size')
```



3. Bar Plot

```
In [11]: sns.barplot(data=tips,x='day',y='total_bill', hue ='sex',palette ='muted')  
plt.title("Barplot of total bill by day")
```

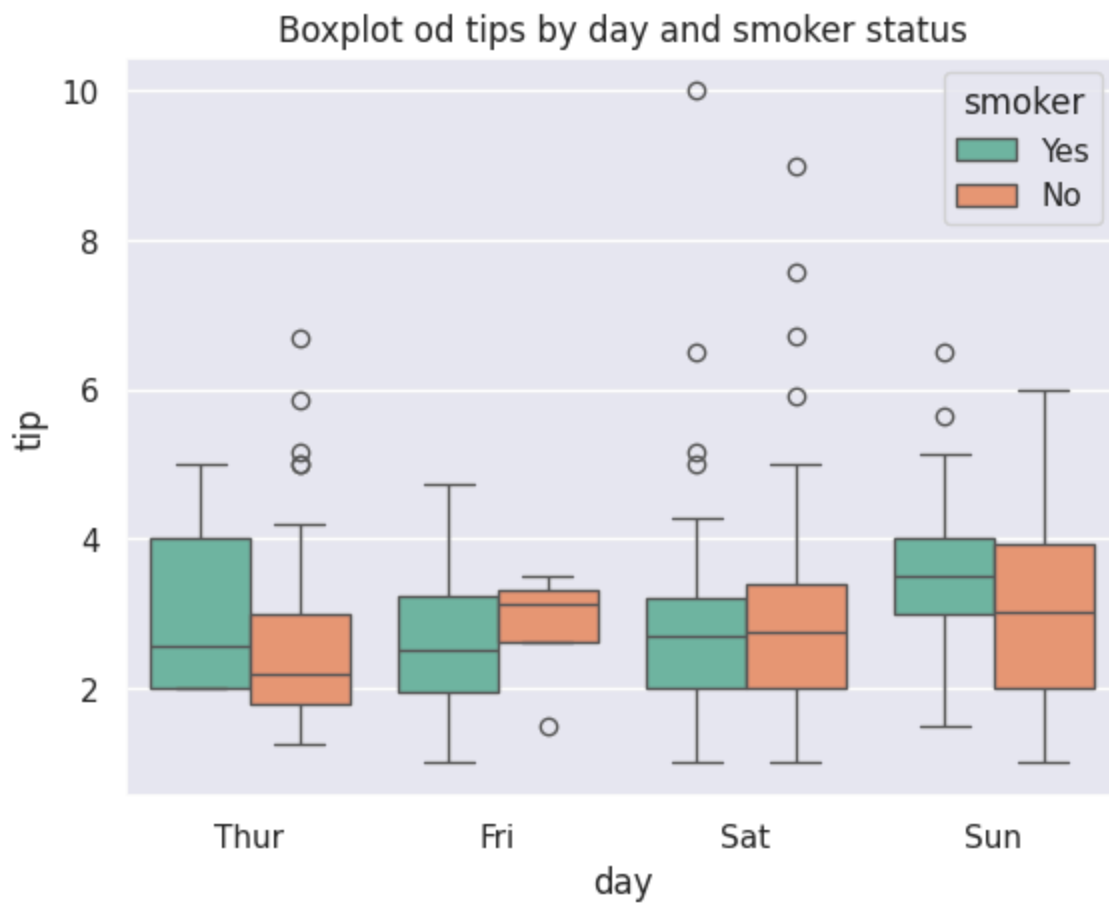
```
Out[11]: Text(0.5, 1.0, 'Barplot of total bill by day')
```



4. BoxPlot

```
In [12]: sns.boxplot(data=tips,x='day',y='tip',hue='smoker',palette='Set2')  
plt.title("Boxplot od tips by day and smoker status")
```

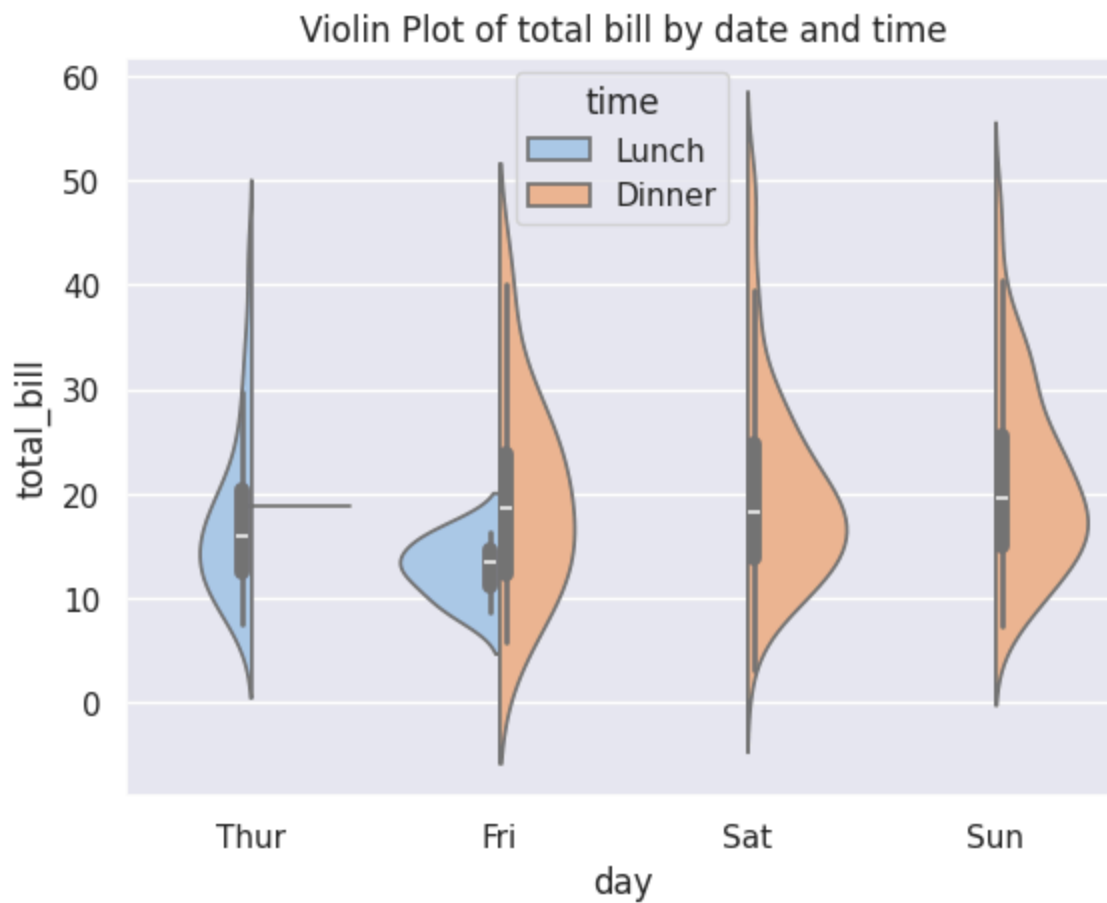
```
Out[12]: Text(0.5, 1.0, 'Boxplot od tips by day and smoker status')
```



5. Violin Plot

```
In [14]: sns.violinplot(data=tips,x='day',y='total_bill',hue='time',split=True,palette=
plt.title("Violin Plot of total bill by date and time")
```

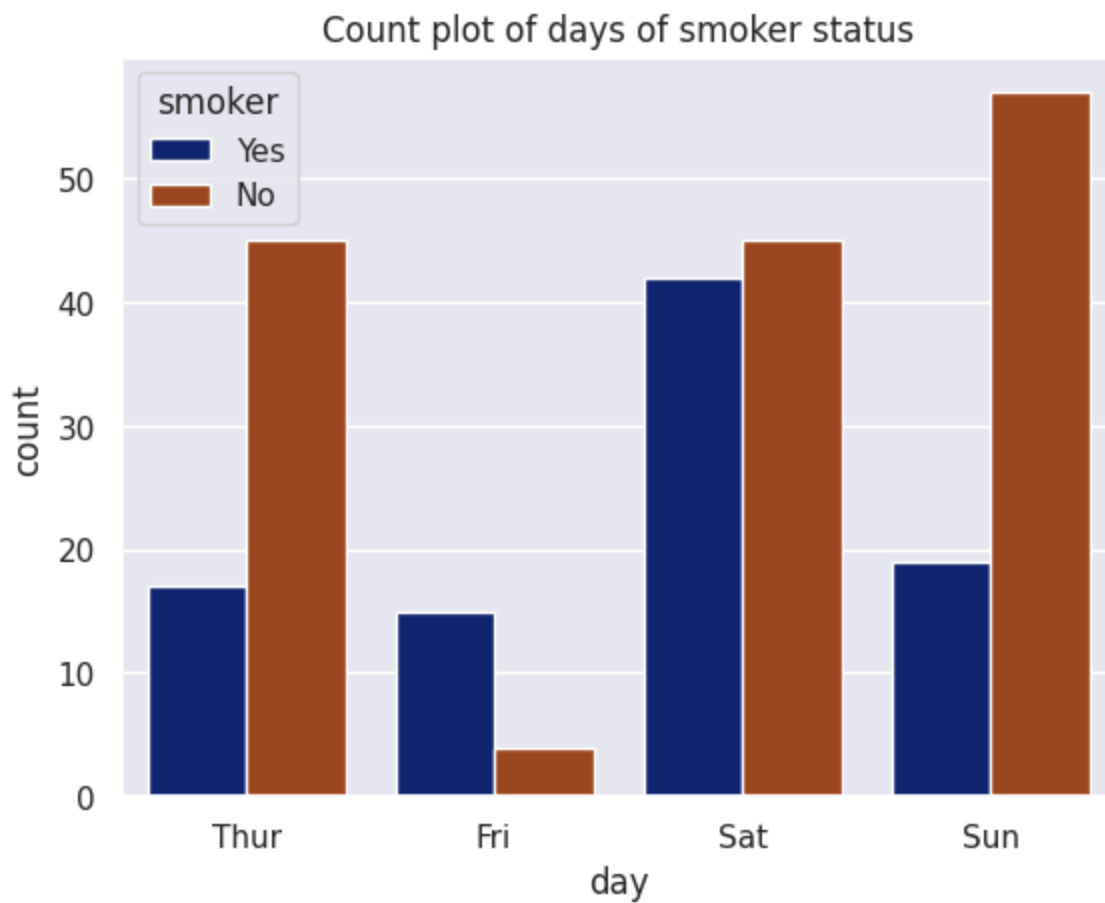
```
Out[14]: Text(0.5, 1.0, 'Violin Plot of total bill by date and time')
```



6. Count Plot

```
In [15]: sns.countplot(data=tips,x='day',hue='smoker',palette='dark')  
plt.title("Count plot of days of smoker status")
```

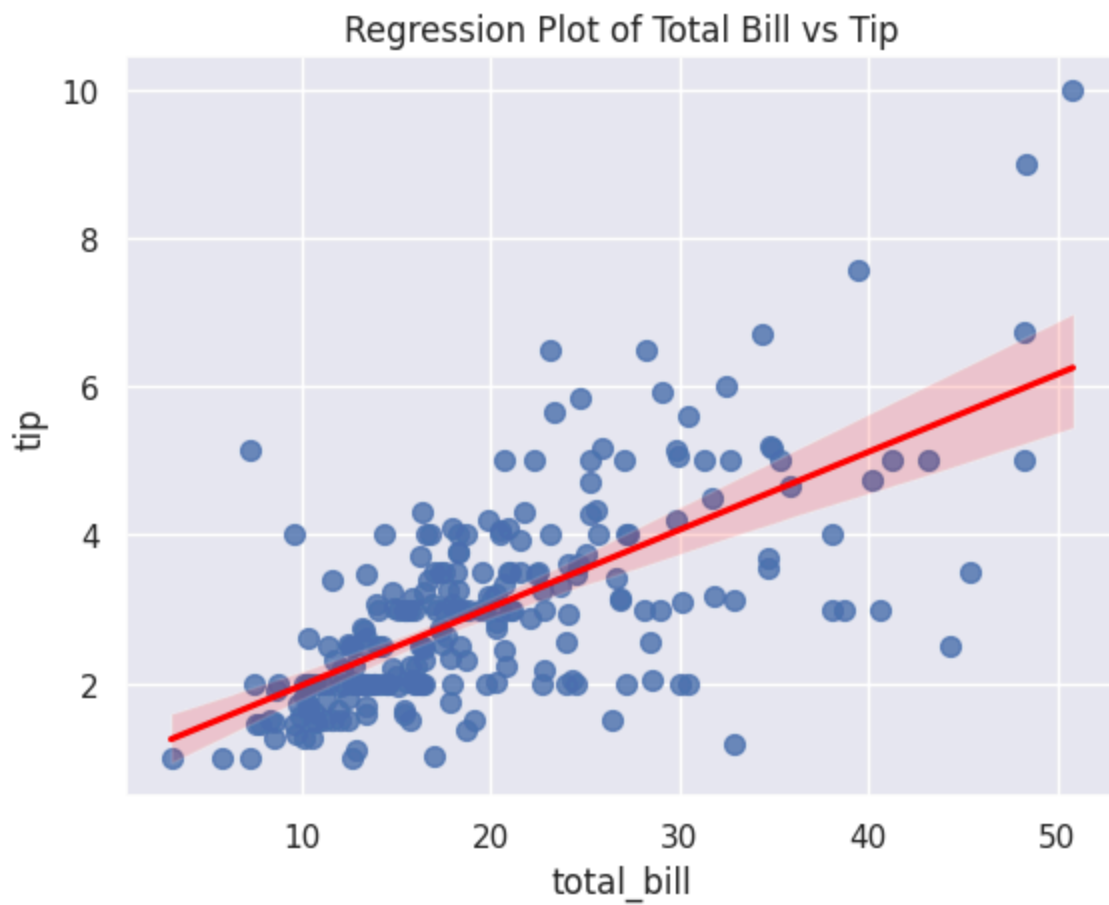
```
Out[15]: Text(0.5, 1.0, 'Count plot of days of smoker status')
```



7. Regression Plot

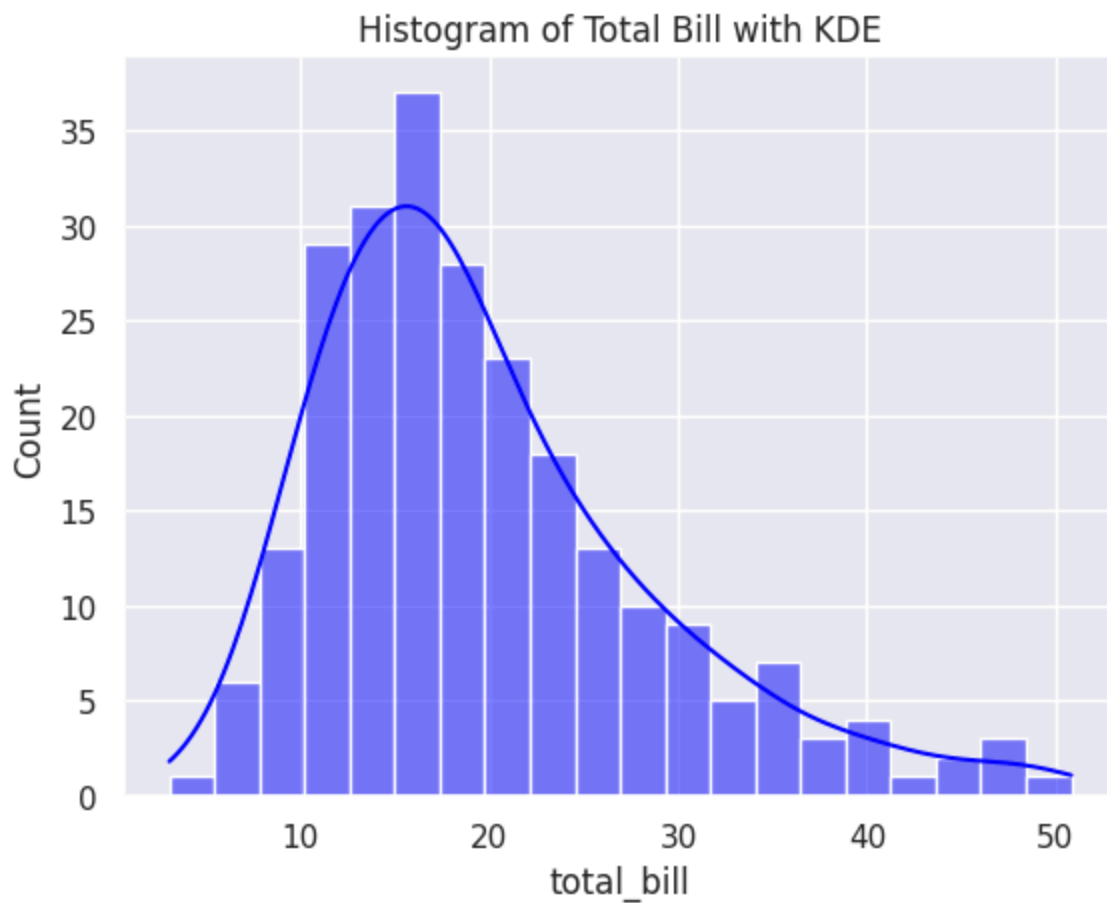
```
In [16]: sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50},  
                    line_kws={'color':'red'})  
plt.title("Regression Plot of Total Bill vs Tip")
```

Out[16]: Text(0.5, 1.0, 'Regression Plot of Total Bill vs Tip')



8. Histogram of total bill with KDE

```
In [17]: sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue')  
plt.title("Histogram of Total Bill with KDE")  
plt.show()
```



9. pairplot

```
In [18]: sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"],  
palette='husl')  
  
plt.suptitle("pair plot: numeric variables by gender", y=1.02)
```

```
Out[18]: Text(0.5, 1.02, 'pair plot: numeric variables by gender')
```

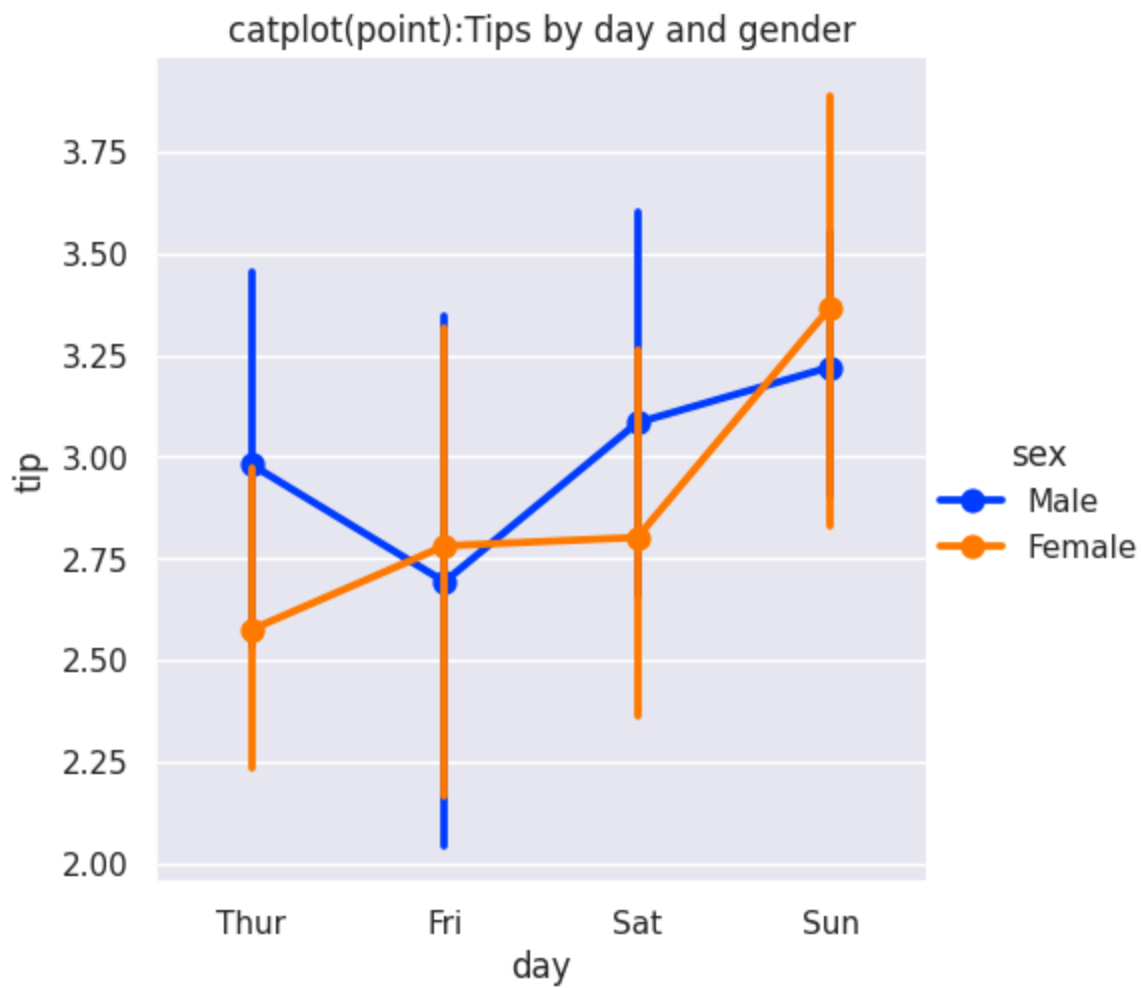


10 catplot

```
In [19]: sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point',
palette='bright')

plt.title("catplot(point):Tips by day and gender")
```

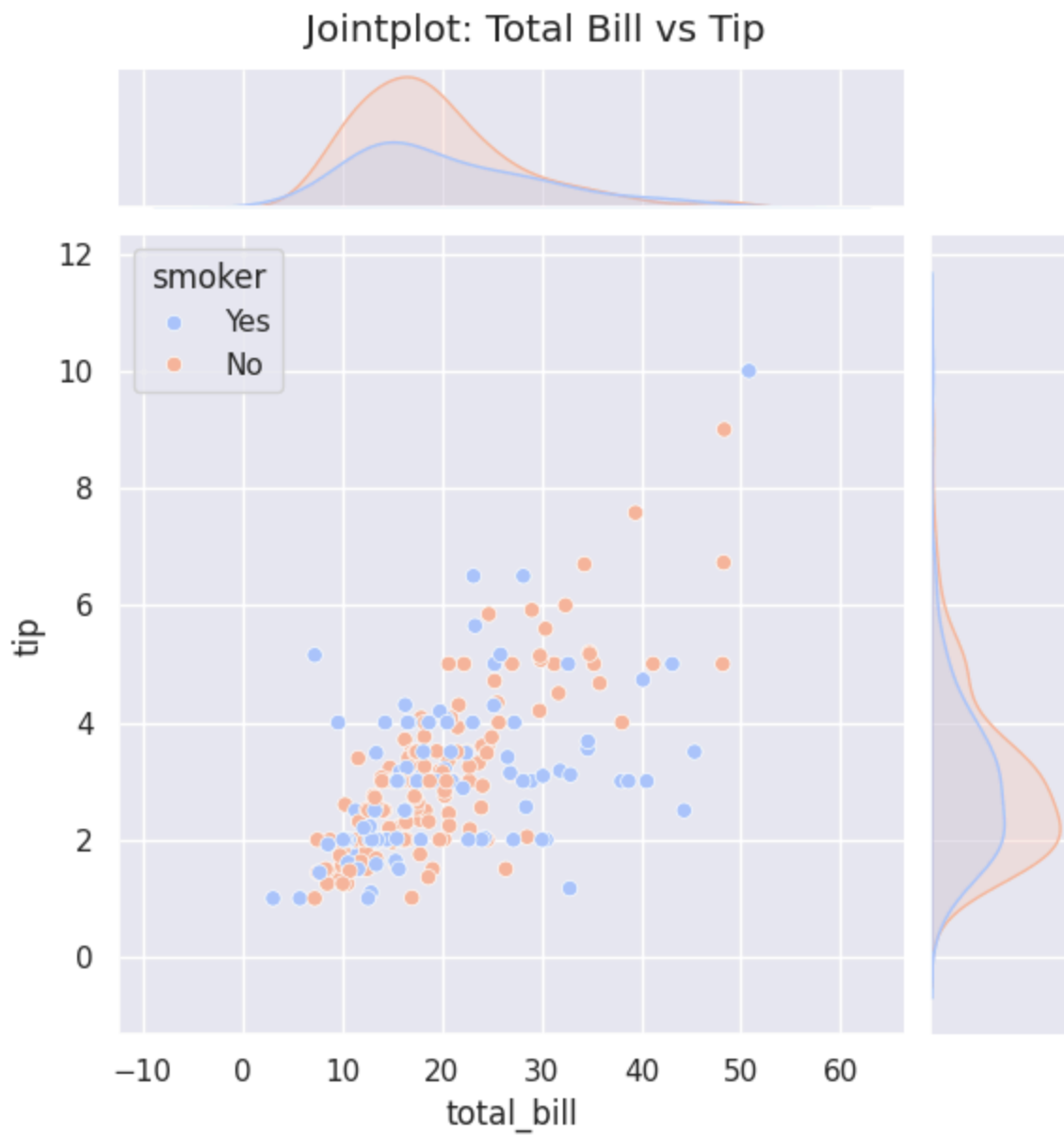
Out[19]: Text(0.5, 1.0, 'catplot(point):Tips by day and gender')



11. jointplot

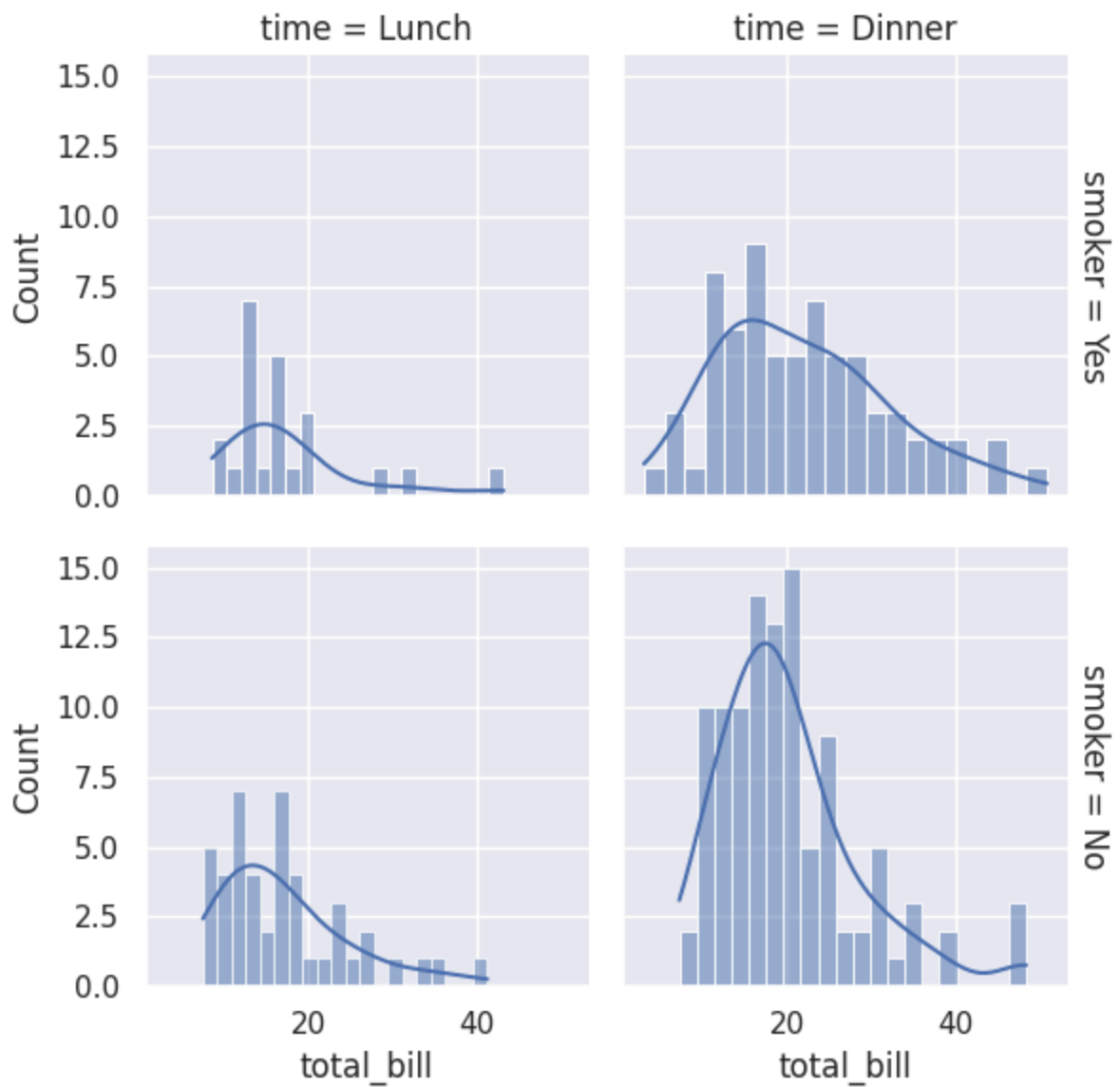
```
In [20]: sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter',  
hue='smoker', color='purple', palette='coolwarm')  
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)
```

Out[20]: Text(0.5, 1.02, 'Jointplot: Total Bill vs Tip')



12. Facetgrid

```
In [21]: g = sns.FacetGrid(tips, col='time', row='smoker',  
margin_titles=True).map(sns.histplot, 'total_bill', bins=20,  
kde=True).add_legend()
```



```
In [22]: plt.figure(figsize=(6,4))
```

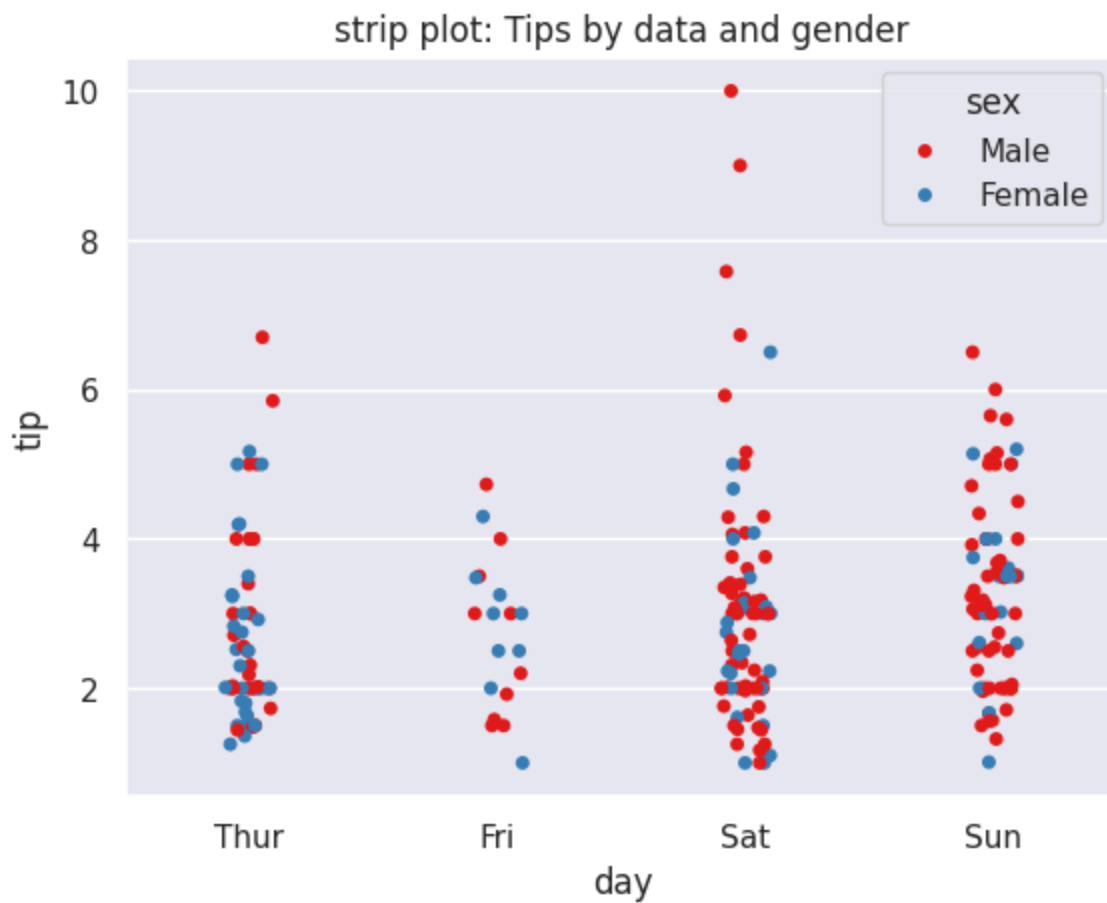
```
Out[22]: <Figure size 600x400 with 0 Axes>
```

```
<Figure size 600x400 with 0 Axes>
```

13. Strip plot

```
In [23]: sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True,
palette='Set1')
plt.title("strip plot: Tips by data and gender")
```

```
Out[23]: Text(0.5, 1.0, 'strip plot: Tips by data and gender')
```



14. KDE PLOT

```
In [24]: sns.kdeplot(data=tips, x='total_bill', hue='sex', fill=True, palette='tab10')
plt.title("kde plot:Total bill density by gender")
plt.show()
```

